

BUTTE MOONEY, MT, USA

WMO: 727740

Lat: 45.965N

Lon: 112.501W

Elev: 1678

StdP: 82.72

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24135

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.7	-22.6	-29.3	0.3	-26.1	-25.5	0.5	-22.3	10.2	0.1	8.8	-0.4	1.2	160	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	31.2	14.1	29.5	13.6	27.8	13.3	15.7	26.4	14.9	25.6	14.1	24.7	4.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	11.1	17.1	11.2	10.2	16.8	9.9	9.3	16.5	49.7	26.3	47.0	25.6	44.6	24.8	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.1	7.2	DB	-33.0	33.6	4.2	1.5	-36.0	34.6	-38.5	35.5	-40.8	36.3	-43.9	37.4	(4)
(5)			WB	-32.2	17.7	3.3	2.5	-34.5	19.5	-36.4	21.0	-38.3	22.4	-40.7	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-6.4	-5.5	-0.3	3.7	8.5	13.1	18.0	16.6	11.5	4.9	-1.8	-7.2	(6)	
	DBStd	9.93	6.60	6.43	5.22	4.17	4.05	3.88	2.85	2.85	4.23	4.67	6.17	6.44	(7)	
	HDD10.0	2632	509	433	318	192	78	15	0	1	32	166	354	533	(8)	
	HDD18.3	5039	767	667	576	438	305	161	41	66	206	418	604	792	(9)	
	CDD10.0	681	0	0	0	3	32	108	247	206	78	7	0	0	(10)	
	CDD18.3	47	0	0	0	0	4	29	12	1	0	0	0	0	(11)	
	CDH23.3	1902	0	0	0	0	26	191	901	623	156	6	0	0	(12)	
	CDH26.7	597	0	0	0	0	2	48	325	189	32	1	0	0	(13)	
(14) Wind	WSAvg	2.6	2.0	2.3	2.8	3.2	3.1	3.0	2.8	2.7	2.5	2.5	2.2	2.0	(14)	
(15) Precipitation	PrecAvg	311	14	10	19	26	47	57	32	35	29	18	13	11	(15)	
	PrecMax	507	42	23	47	65	122	147	106	79	80	78	34	28	(16)	
	PrecMin	207	1	1	5	3	12	7	2	4	2	0	4	2	(17)	
	PrecStd	79	10	6	9	15	24	33	26	21	20	16	7	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	10.6	17.1	21.8	26.2	30.9	33.9	32.4	30.0	23.8	16.2	8.4	(19)	
		MCWB	3.4	3.6	6.2	9.0	11.5	13.9	15.5	14.6	13.1	10.3	6.5	3.4	(20)	
	2%	DB	6.3	7.5	13.1	18.7	23.3	27.9	31.9	30.4	27.3	20.1	12.7	6.0	(21)	
		MCWB	2.0	2.4	4.8	7.6	10.7	13.2	14.6	13.8	12.5	8.7	5.2	2.1	(22)	
	5%	DB	4.5	5.5	10.7	15.7	21.1	25.5	30.1	28.9	25.1	17.1	9.6	4.0	(23)	
		MCWB	0.8	1.4	3.8	6.5	9.8	12.6	14.0	13.3	11.5	7.8	4.2	0.8	(24)	
	10%	DB	2.6	3.7	8.0	12.6	18.2	22.9	28.4	27.2	22.5	13.9	7.1	2.3	(25)	
		MCWB	-0.4	0.2	2.6	5.1	8.9	11.6	13.6	13.1	10.7	6.5	2.7	-0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.5	6.8	9.8	13.2	15.4	17.1	16.5	15.2	11.0	7.8	3.9	(27)	
		MCDB	8.2	9.2	14.6	19.9	21.9	26.5	28.7	27.3	23.3	21.3	15.3	7.5	(28)	
	2%	WB	2.3	2.9	5.3	8.2	11.6	14.1	15.9	15.5	13.6	9.6	5.7	2.4	(29)	
		MCDB	5.7	6.8	12.4	17.2	20.7	24.6	27.1	25.8	23.3	18.3	11.3	5.6	(30)	
	5%	WB	1.0	1.7	4.1	6.8	10.5	13.2	15.2	14.6	12.5	8.4	4.4	1.0	(31)	
		MCDB	4.1	5.2	9.8	14.8	19.1	23.1	26.3	24.8	22.5	15.9	9.4	3.8	(32)	
	10%	WB	-0.1	0.4	2.8	5.5	9.4	12.3	14.5	13.8	11.4	6.9	2.9	-0.4	(33)	
		MCDB	2.6	3.5	7.7	12.0	17.4	21.4	25.5	24.1	21.0	13.4	6.8	2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.7	12.7	13.0	13.8	15.1	18.9	18.9	16.9	14.2	12.8	11.7	(35)	
		MCDBR	12.3	13.4	16.2	19.5	20.0	20.5	22.1	22.5	22.2	20.7	16.1	10.9	(36)	
	5% WB	MCWBR	8.8	9.0	9.1	9.7	9.0	7.9	8.0	8.8	9.5	10.8	9.7	7.9	(37)	
		MCWBR	11.3	12.4	14.9	17.8	17.3	17.9	18.9	19.1	19.2	18.5	14.9	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.235	0.244	0.259	0.286	0.294	0.304	0.310	0.319	0.290	0.272	0.246	0.233	(40)	
		taud	2.524	2.529	2.588	2.512	2.559	2.564	2.559	2.496	2.571	2.604	2.596	2.550	(41)	
	Ebn at Noon		910	964	989	978	974	961	950	930	938	915	889	876	(42)	
		Edh at Noon	61	75	83	98	97	98	97	99	84	70	57	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.53	2.48	3.75	4.93	5.68	6.56	7.17	6.11	4.56	2.87	1.70	1.22	(44)	
	RadStd		0.14	0.20	0.22	0.34	0.50	0.50	0.41	0.32	0.40	0.23	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.58	-1.16	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	-0.32	-0.64	N/A	N/A	N/A	N/A

Nomenclature: See separate page